

Predictive Forecasting of Care Load & Placement Demand

1. Introduction:

The increasing demand for care facilities requires proactive planning and data-driven decision-making. This project develops a predictive forecasting system to estimate future care load and placement demand using historical intake and discharge trends.

The objective is to:

- Forecast future HHS care load
- Detect system pressure early
- Identify capacity risks
- Support planning and resource allocation

The system is implemented using Python, Streamlit, and machine learning techniques.

2. Dataset Description:

The dataset used is:

HHS Unaccompanied Alien Children Program Dataset

It contains daily records including:

- CBP Apprehended
- Children in CBP Custody
- Children Transferred to HHS
- Children in HHS Care
- Children Discharged from HHS Care
- Date

The dataset spans multiple years of daily observations, enabling time-series modeling.

3. Data Preparation:

- Removed commas and special characters
- Converted numeric columns to proper numeric types
- Converted date column to datetime format
- Sorted data chronologically
- Handled missing values using interpolation
- Ensured daily frequency consistency

The following predictive features were created:

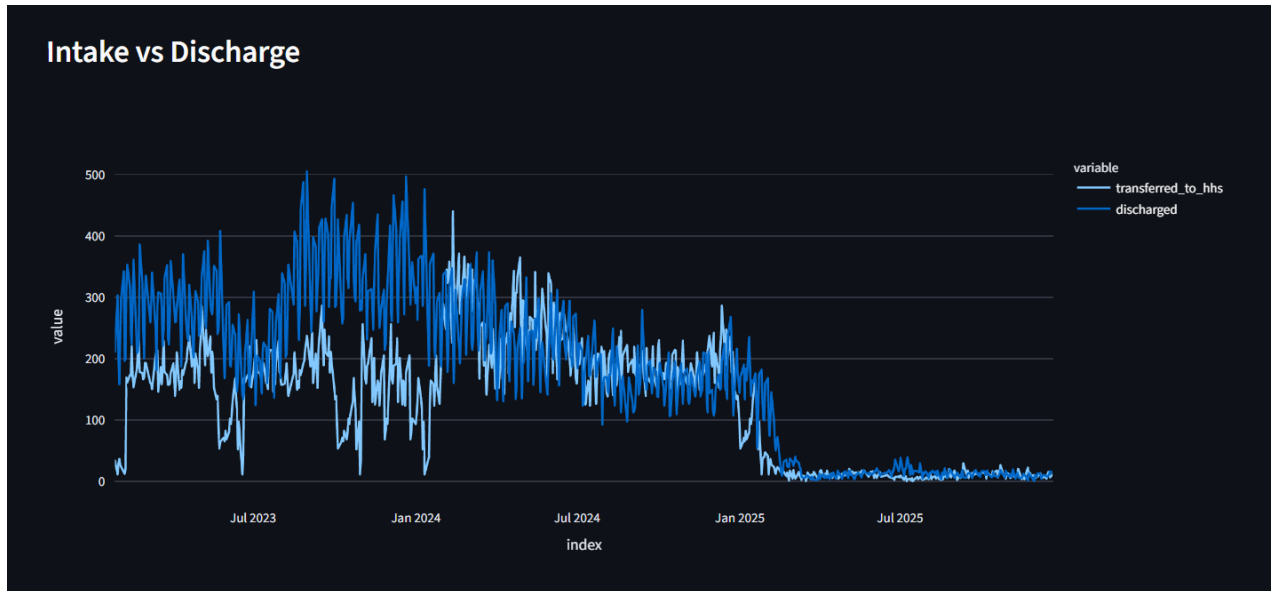
- Lag features (lag_1, lag_7, lag_14)
- Rolling mean (7-day, 14-day)

- Rolling standard deviation
- Net pressure = Intake – Discharge
- Calendar features (optional)

These features help capture trend, seasonality, and short-term momentum.

4. Exploratory Data Analysis (EDA):

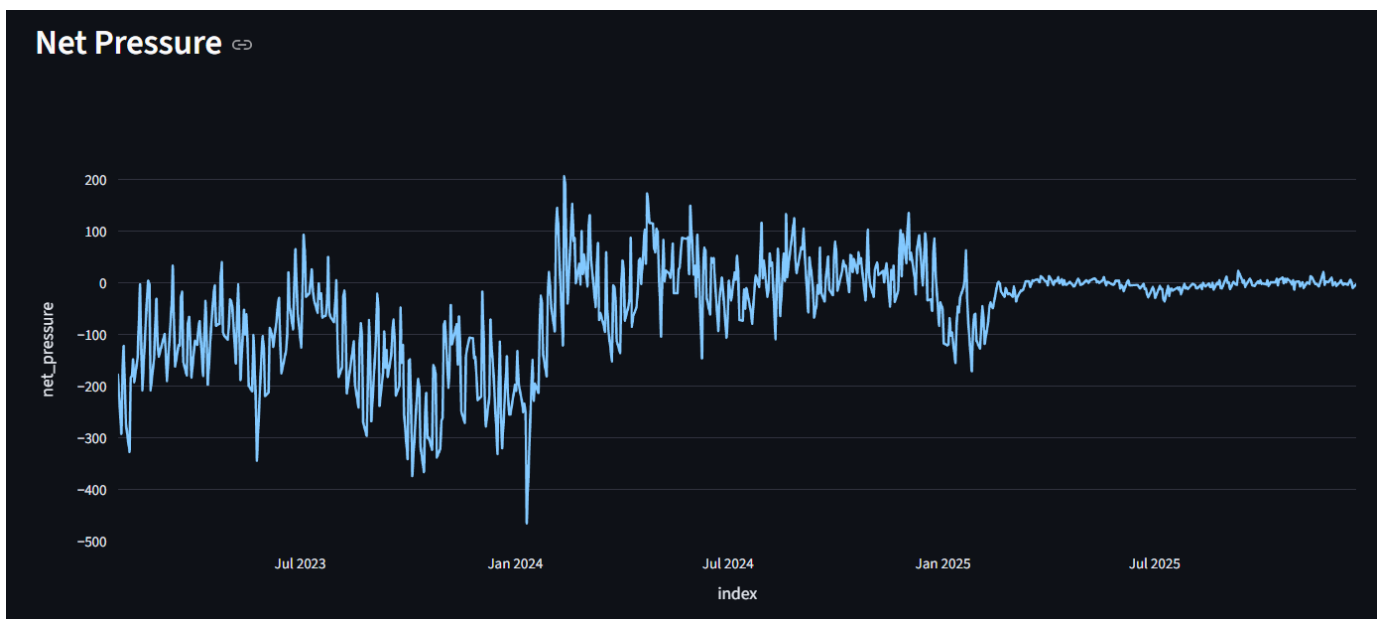
Figure 1: Intake vs Discharge Trend



Observation:

- Intake and discharge fluctuate significantly.
- Periods where intake exceeds discharge increase system pressure.
- A structural shift is observed around early 2025, where flow values decline sharply. This may indicate policy or reporting changes.

Figure 2: Net Pressure Indicator



Net Pressure = Transferred to HHS – Discharged

Observation:

- Positive values indicate system strain.
- Negative values indicate discharge exceeding intake.
- Pressure volatility directly impacts care load forecasting.

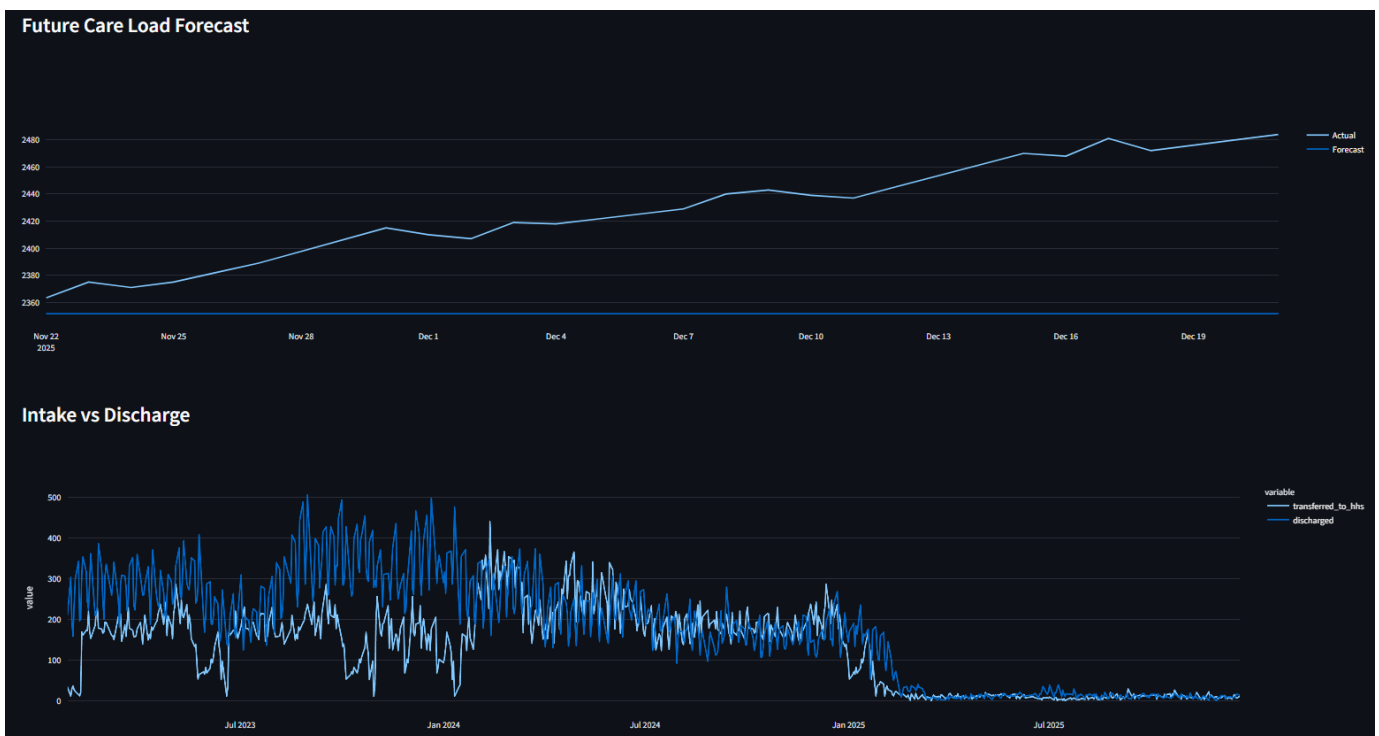
5. Forecasting Models Implemented:

Three models were implemented and compared.

Naive Baseline Model

- Predicts future values using the last observed value.
- Serves as performance benchmark.

Figure 3: Naive Forecast



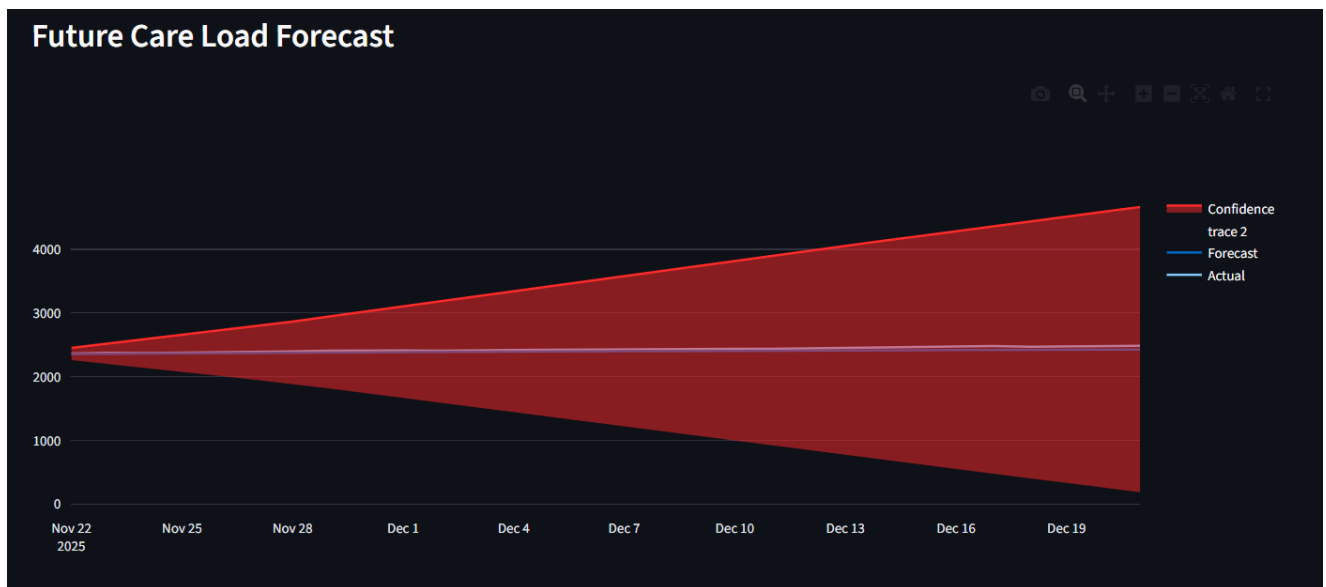
Observation:

- Produces flat-line predictions.
- Fails to adapt to trend changes.
- Highest forecasting error.

SARIMA Model (Statistical Approach)

- Seasonal ARIMA model used for time-series forecasting.
- Captures trend and seasonality.
- Provides confidence intervals.

Figure 4: SARIMA Forecast



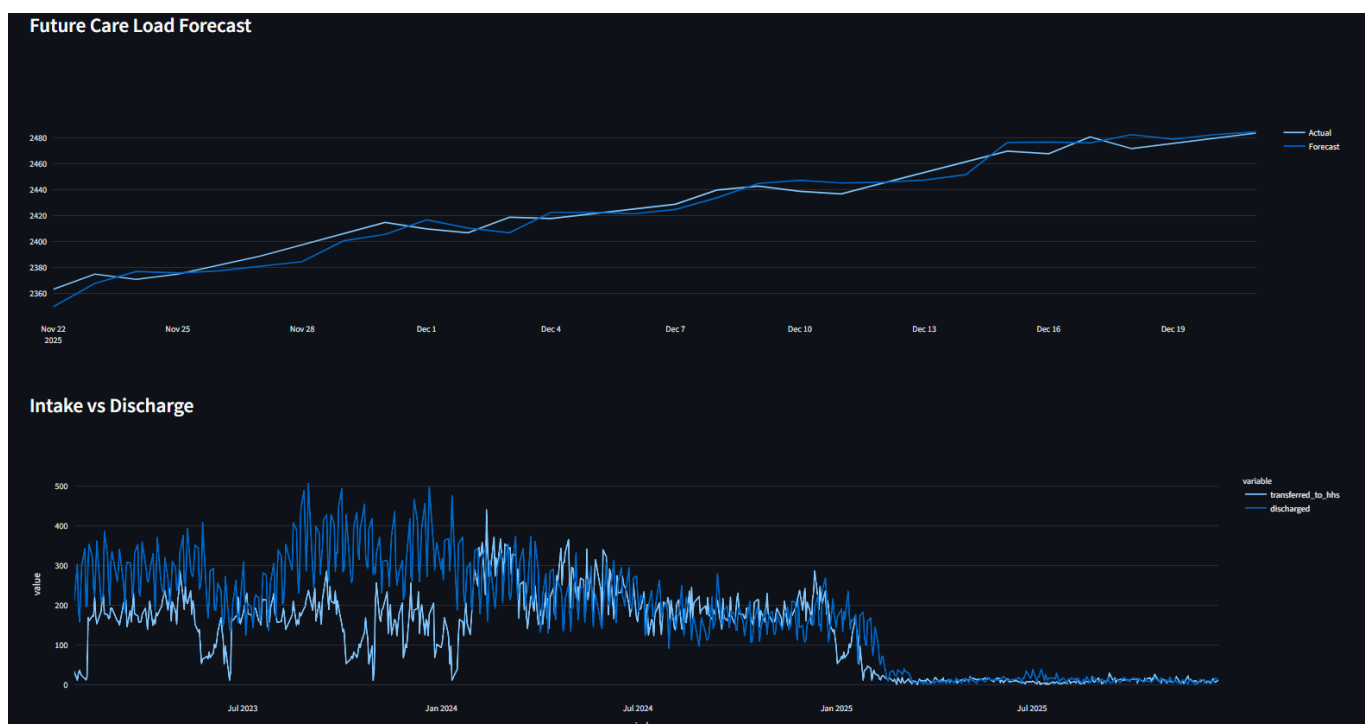
Observation:

- Captures general upward trend.
- Confidence interval widens over time.
- Performs significantly better than Naive model.

Random Forest Regressor (Machine Learning Model)

- Uses lag and rolling features.
- Captures nonlinear relationships.
- Trained using historical data split into train/test sets.

Figure 5: Random Forest Forecast



Observation:

- Closely tracks actual values.
- Very low deviation.
- Most stable predictions among all models.

6. Model Performance Comparison:

Figure 6: Model Comparison Table

Model Comparison				
	Model	MAE	RMSE	MAPE
0	SARIMA	32.78	36.54	1.34
1	Random Forest	6.13	7.09	0.25
2	Naive	76.84	84.72	3.14

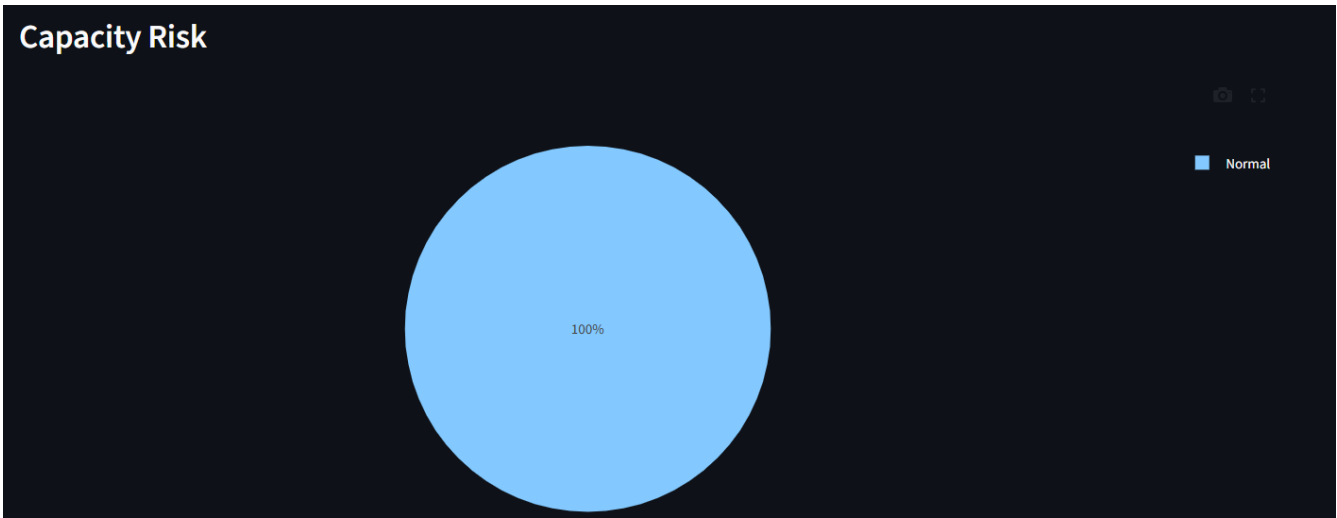
Random Forest reduced MAPE by more than 80% compared to SARIMA and significantly outperformed the Naive baseline.

7. Capacity Risk Monitoring:

Capacity thresholds defined as:

- Normal
- Warning (>12,000)
- Critical (>15,000)

Figure 7: Capacity Risk Pie Chart



Observation:

- No critical breach observed in selected timeframe.
- Dashboard supports early warning monitoring.

8. Dashboard Implementation:

The dashboard was built using:

- Streamlit (Frontend UI)
- Plotly (Interactive visualization)

- Scikit-learn (Machine Learning)
- Statsmodels (SARIMA)

Features include:

- Forecast horizon selection
- Model selection
- Date range filtering
- KPI display (MAE, RMSE, MAPE)
- Intake vs discharge trends
- Net pressure visualization
- Capacity risk monitoring
- Model comparison table

9. Result Insights:

- Random Forest is the most accurate forecasting model.
- Intake-discharge imbalance drives system pressure.
- Forecast uncertainty increases with longer horizons.
- Machine learning captures nonlinear behavior better than statistical models.
- Structural shift in early 2025 may affect long-term model stability.

10. Business Recommendations:

- Use Random Forest for operational forecasting.
- Monitor net pressure daily.
- Scale staff and resources before projected surges.
- Investigate structural break observed in 2025 data.
- Use dashboard as planning support tool.

11. Conclusion:

This project successfully developed a predictive forecasting system for care load management. Through statistical and machine learning comparison, Random Forest demonstrated superior performance and reliability.

The dashboard provides a decision-support system for proactive planning, risk monitoring, and data-driven capacity management.

The system can be extended with:

- Real-time API integration
- Advanced hyperparameter tuning
- External policy or migration indicators